

The Overly Complicated Door Alarm Design Specification

The Overly Complicated Door Alarm is the latest and greatest in intrusion detection technology. A tilt switch placed in front of a door leads to a series of alarms going off to alert the user of any attempted intrusion.

Materials: The Overly Complicated Door Alarm requires the following parts and materials:

Wood: (1) 4'x4' sheet of plywood, (2) 12" lengths of 2x4, (1) 4" length of 2x4, (1) block, approx. 2"x2"x1", (1) 1" length of ½" dowel

Electrical: 10' each of black and red 22 gauge wire, (1) roll black electrical tape, (4) ring terminals, (1) Tamiya 70168 double gearbox kit, (2) AA battery boxes: 1) 4 battery box, 1) 2 battery box, (1) Adafruit tiny breadboard, (6) AA batteries, (10) 90db window alarms, (1) 4' length of aluminum foil tape, 4" heat shrink tubing

Robot: (1) MakeIt Robotics starter kit, (1) Arduino Uno board (1) mousetrap, (1) zip tie, (2) thumb tacks, (8) AA batteries

Hardware & misc.: (9) 1.5" wood screws, (1) ¾" length flange head screw, (1) 2' length nylon string, (2) self-tapping hex washer screws, (2) 10-32x ¾" round head screws, (1) 1" steel ball bearing, (1) ¼" x 1½" x 2½" u-bolt, (1) ¼" x 1½" x 2" u-bolt, white paint to provide contrast to the black line

The system consists of a tilt switch which sits in front of the desired door. Opening the door causes it to tip over, causing the steel ball bearing inside to close a circuit connected to a gearbox. The gearbox acts as a winch, pulling the block to which it is attached through a channel towards an awaiting line-following robot. Both the block and the channel in which it sits are lined with foil tape and are integrated into the electrical circuit of the gearbox, so that when the block clears the channel, the circuit is broken, turning off the gearbox. The line-following robot waits in the path of the block with a mousetrap zip-tied to the back. The mousetrap is tripped by a screw protruding from the front of the block. Wires are connected to the mousetrap from the robot such that when it snaps shut, the wires are connected, closing the circuit to power up the robot. The mousetrap is tripped by a screw protruding from the front of the block. Once the robot

turns on, it travels along a black line which has magnetic window alarms placed along the length of it. As the robot passes by the window alarms, a bar at the front of it sweeps aside the magnet portion of the alarm, activating the siren. As it continues along the line, more alarms are tripped, creating an increasingly loud indication of intrusion into the home.

Prior to delivering the final product, the following components should be tested:

Tilt switch: test to ensure circuit remains open while switch remains upright, circuit is closed when switch has tipped over, and switch can stand on its own until knocked over by the door.

Gearbox: Test gearbox to ensure motors function, ensure line wraps around axles for proper winch functioning, and ensure block maintains contact with channel through entire length of the channel sufficiently that the circuit remains closed and gearbox continues running.

Robot: Test functioning of mousetrap when tripped by the block; i.e., mousetrap bar snaps all the way closed free of any obstacles, completing circuit. Test functioning of robot to verify it follows the black line.

Alarms: test all alarms to ensure: a) siren remains silent while magnet is attached, and b) siren activates when magnet is removed

Project Management: This project should take approximately 3 hours to fully assemble. No status update meetings are necessary unless unforeseen delays occur.

The following items should be included with the product:

Spare parts: extra foil tape should be included to enable end user to close any gaps between the block and the channel as needed. Any other small parts will be provided in accordance with company warranty policy.

Documentation: (1) copy of assembly instructions with diagrams, (1) parts list, (1) copy of test results initialed by q.c. testor.

Drawings: drawings of all parts designed and manufactured in-house should be included.

Software: Programming code for the line following robot can be found online at:

<http://blog.radioshack.com/2014/05/radioshack-make-magazine-launch-make-robotics/>

Points of Contact: Any questions or issues pertaining to the Overly Complicated Door Alarm may be directed to: